A Jordan obstruction and generic irreducibility for Long–Moody braid representations

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Abstract

We resolve the Jones–Temperley–Lieb Jordan-obstruction clause of Negami's Problem 6.8 and extend its generic irreducibility result to $n = 5$. For every untwisted endpoint-sector seed of dimension at least two, with $n \geq 3$ and $\ell \geq 4$, an explicit four-dimensional block contains a nontrivial length-two Jordan chain. The chain survives every admissible middle-convolution quotient, so both braid images are unbounded and admit no positive-definite invariant Hermitian form. For the separate character-twisted Burau family, an exact minor of order 893 proves generic irreducibility of the twenty-dimensional summand $\mathcal{R}_5$ in the positive parameter region. We construct an invariant $10 + 10$ splitting along the entire admissible diagonal $\tau = q$ and prove that every ambient minor of order 893 is divisible by $(\tau - q)^3$. Exact cyclotomic certificates establish irreducibility at 95 off-diagonal points and identify the two irreducible constituents at seven diagonal points.

1 The construction and the two questions

Negami constructs braid representations from character-twisted Burau seeds and asks in [1, Problem 6.8] for the remaining irreducibility locus, the image closure on irreducible sectors, and an exact Jordan obstruction for the Jones–Temperley–Lieb seeds of [1, Remark 6.7(b)]. We prove the Jordan obstruction uniformly in the endpoint sector, including its survival on the convolution quotient. For the Burau family we extend [1, Proposition 5.10] from $n = 3, 4$ to $n = 5$ and determine an explicit reducibility curve. These are two different seed families.

We use the conventions of [1, Definition 1.3 and Definitions 1.4–1.7]. A seed consists of invertible operators $g_1, \ldots, g_n$ and $s_1, \ldots, s_{n-1}$ on a vector space V satisfying the seed relations (the braid and Artin compatibility relations of Definition 1.3). The ambient Long–Moody module is $U = V^{\oplus n}$. Its braid generator S_i acts by s_i on every slot other than $i, i+1$, and

$$(S_i v)_i = s_i g_i v_{i+1}, \qquad (S_i v)_{i+1} = s_i(v_i + (I - g_{i+1})v_{i+1}). \tag{1}$$

The middle-convolution module, for $\lambda \in \mathbb{C} \setminus \{0, 1\}$, is

$$Q_\lambda = U/(K_{\mathrm{seed}} + L_\lambda), \qquad K_{\mathrm{seed}} = \bigoplus_{j=1}^{n} \ker(g_j - I),$$

where, with $P_n = g_1 \cdots g_n$,

$$L_\lambda = \{z : z_j = g_{j+1} z_{j+1} \ (1 \leq j < n), \quad z_n = \lambda P_n z_n\}. \tag{2}$$

These subspaces are braid invariant. The operators in (1) are independent of λ [1, Proposition 2.3].

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2 The endpoint-sector Jordan obstruction

Fix integers $n \geq 3$ and $\ell \geq 4$. An endpoint sector V has an orthonormal basis indexed by paths $p = (p_0, \ldots, p_{n+1})$ on $A_{\ell-1}$, with vertices $1, \ldots, \ell-1$, $p_0 = 1$, and a fixed terminal vertex. Consecutive vertices differ by one. Set

$$\theta = \pi/\ell, \qquad \mu_c = \sin(c\theta), \qquad \delta = 2\cos\theta, \qquad \alpha = ie^{-i\theta/2}, \qquad b = -\alpha^{-3}.$$

The operator E_j, $0 \leq j < n$, changes vertex $j+1$ only. Its entry from a path r to a path p is zero unless the paths agree away from $j+1$ and $p_j = p_{j+2}$; otherwise

$$(E_j)_{p,r} = \frac{\sqrt{\mu_{p_{j+1}}\mu_{r_{j+1}}}}{\mu_{p_j}}. \tag{3}$$

The untwisted normalization of [1, Remark 6.7(b)] is

$$\rho_j = \alpha I + \alpha^{-1}E_j, \qquad s_j = \rho_j, \qquad g_1 = \rho_0^2, \qquad g_{j+1} = \rho_j g_j \rho_j^{-1}.$$

Theorem 1. *For every endpoint sector V of dimension at least two, $K_{\mathrm{seed}} = 0$, and there are $w, v \in U$ such that*

$$(S_1 - bI)w = v \neq 0, \qquad (S_1 - bI)v = 0.$$

The images of w, v give the same nontrivial Jordan chain on Q_λ for every $\lambda \in \mathbb{C} \setminus \{0, 1\}$. Both the ambient and quotient braid images are unbounded and preserve no positive-definite Hermitian form.

Proof. Put $\mu_0 = \mu_\ell = 0$. The sine addition formula gives $\mu_{c-1} + \mu_{c+1} = \delta\mu_c$, also at the boundary vertices. Every nonzero local block of E_j is uu^{T}/μ_c, where $u_d = \sqrt{\mu_d}$ for neighbors d of c. Thus $E_j^2 = \delta E_j$. Since $\delta = -(\alpha^2 + \alpha^{-2})$, we obtain

$$(\rho_j - \alpha I)(\rho_j + \alpha^{-3}I) = 0, \qquad (g_j - \alpha^2 I)(g_j - \alpha^{-6}I) = 0.$$

The second identity follows first for g_1 by squaring, then for all g_j by conjugation. The seed operators are invertible because the two roots $\alpha, -\alpha^{-3}$ are nonzero. Now $\alpha^2 = -e^{-i\theta}$ and $\alpha^6 = -e^{-3i\theta}$ have positive imaginary parts for $0 < \theta \leq \pi/4$. Consequently neither α^2 nor α^{-6} is one. Applying the polynomial identity to a fixed vector of g_j shows that it is zero, proving $K_{\mathrm{seed}} = 0$.

There is at most one strictly ascending path. Since the sector has at least two paths, some path has a downward step. At the first such step, delete the two-edge detour $c-1, c, c-1$ and prepend the loop $1, 2, 1$. This preserves length and endpoint and produces a path beginning $1, 2, 1, 2$. Here the fourth vertex is present because $n + 1 \geq 4$, and a step after vertex 1 must go to 2. Replacing the third vertex by 3 gives a second legal path beginning $1, 2, 3, 2$, since $\ell \geq 4$. Denote these paths by p, p'; their tails agree. On their span,

$$E_0 = \begin{pmatrix} \delta & 0 \\ 0 & 0 \end{pmatrix}, \qquad E_1 = \frac{1}{\mu_2}\begin{pmatrix} \mu_1 & \sqrt{\mu_1\mu_3} \\ \sqrt{\mu_1\mu_3} & \mu_3 \end{pmatrix}.$$

Put $f_0 = e_p$ and $f_1 = \alpha^2 E_1 f_0$. The coefficient of $e_{p'}$ in f_1 is nonzero, so f_0, f_1 are independent. Using $\mu_2 = \delta\mu_1$ gives $E_0 f_1 = \alpha^2 f_0$ and $E_1 f_1 = \delta f_1$. In this basis the first two seed matrices are

$$R = \rho_0 = \begin{pmatrix} -\alpha^{-3} & \alpha \\ 0 & \alpha \end{pmatrix}, \qquad U = \rho_1 = \begin{pmatrix} \alpha & 0 \\ \alpha^{-3} & -\alpha^{-3} \end{pmatrix}, \qquad U^{-1} = \begin{pmatrix} \alpha^{-1} & 0 \\ \alpha^{-1} & -\alpha^3 \end{pmatrix}. \tag{4}$$

Let M be the four-dimensional space consisting of this span in slots $1, 2$ and zero in all other slots. It is S_1 invariant, with

$$T = S_1|_M = \begin{pmatrix} 0 & UG \\ U & U(I-H) \end{pmatrix}, \qquad G = R^2, \qquad H = UGU^{-1}. \tag{5}$$

In the order (f_0, f_1) in slot 1, then (f_0, f_1) in slot 2, put $d = \alpha^6 - 1$ and set

$$w = \begin{pmatrix} -\alpha^5(\alpha^2+1)(\alpha^8-\alpha^6-\alpha^4-\alpha^2+1) \\ \alpha(2\alpha^6+\alpha^4-1) \\ -\alpha^{11}(\alpha^2+1) \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} -\alpha^2(\alpha^4-1)d \\ \alpha^{-2}d \\ 0 \\ d \end{pmatrix}. \tag{6}$$

Multiplying the matrices in (4)–(5) and collecting Laurent powers gives the identities

$$(T + \alpha^{-3}I)w = v, \qquad (T + \alpha^{-3}I)v = 0 \tag{7}$$

for every nonzero α. At the specified phase, $d \neq 0$, so the last coordinate of v is nonzero. Extension by zero to the remaining slots proves the ambient chain.

If $z \in M \cap L_\lambda$, then $z_3 = \cdots = z_n = 0$. The recurrence in (2) gives $z_2 = g_3 z_3 = 0$ and $z_1 = g_2 z_2 = 0$. Hence $M \cap L_\lambda = 0$. Since $K_{\text{seed}} = 0$, the quotient map is injective on M, and (7) descends with $[v] \neq 0$. This argument includes the parameters for which $L_\lambda \neq 0$.

Finally $|b| = 1$, and for every integer $k \geq 1$ the chain gives $S_1^k w = b^k w + k b^{k-1} v$. In any norm,

$$\|S_1^k w\| \geq k\|v\| - \|w\| \longrightarrow \infty.$$

The same inequality holds on the quotient. An invariant positive-definite Hermitian form would make every power an isometry, a contradiction. □

All standard braid generators are conjugate, so each has the same Jordan obstruction. A formalization of the ambient assertion covers $K_{\text{seed}} = 0$ and the existence of the length-two chain with the path conventions above. The quotient and unboundedness conclusions follow from the written argument.

3 The character-twisted Burau summand

For the remainder of the paper set

$$\mathcal{T}_n = \{(a, c) : a > 0,\ c > 0,\ nc + (n+1)a < 1\}, \qquad q = e^{2\pi i a}, \qquad \tau = e^{2\pi i c}.$$

Let $B_j(q)$, $0 \leq j < n$, be the unreduced Burau matrix on $\mathbb{C}^{n+1}$ with local block $\left(\begin{smallmatrix} 1-q & q \\ 1 & 0 \end{smallmatrix}\right)$ at coordinates $j, j+1$ and identity elsewhere. Define

$$s_i = B_i \ (1 \leq i < n), \qquad g_1 = \tau B_0^2, \qquad g_{j+1} = B_j g_j B_j^{-1}.$$

Write $U_0 = (\mathbb{C}^{n+1})^{\oplus n}$ for the resulting ambient module. The reduced seed is $V = \ker \ell_q$, where $\ell_q(x) = \sum_{j=0}^{n} q^j x_j$; its integral Laurent basis is $b_j = e_j - q^j e_0$, $1 \leq j \leq n$. On $V^{\oplus n}$ put

$$D(v) = \sum_{j=1}^{n} (g_j - I)v_j, \qquad \mathcal{R}_n = \ker D.$$

On $\mathcal{T}_n$, $g_1 - I$ is invertible, so D is surjective and $\dim \mathcal{R}_n = n(n-1)$. By [1, Proposition 5.9], the map $(\Psi v)_j = v_j - g_{j+1} v_{j+1}$ identifies $\mathcal{R}_n$ with $V^{\oplus(n-1)}$. Indeed, its kernel is $F(V)$, where

$$F(v) = (g_2 \cdots g_n v, \ldots, g_n v, v), \qquad DF = P_n - I,$$

and $P_n - I$ is invertible on $\mathcal{T}_n$ [1, Theorem 5.3 and Remark 5.8]. In these coordinates the generator is

$$(A_i u)_j = \begin{cases} s_i(u_{i-1} + g_i u_i), & j = i-1, \\ -s_i g_i u_i, & j = i, \\ s_i(u_i + u_{i+1}), & j = i+1, \\ s_i u_j, & \text{otherwise}, \end{cases} \tag{8}$$

with nonexistent endpoint rows omitted and all seed operators restricted to V. For $n = 5$, the ambient and reduced dimensions are 30 and 20.

4 A nonzero Laurent minor and generic irreducibility

For row-major vectorization of a $d \times d$ matrix X, the commutant system for generators $M_1, \dots, M_{n-1}$ is

$$\mathcal{C}(M) = \begin{pmatrix} M_1 \otimes I_d - I_d \otimes M_1^{\mathsf{T}} \\ \vdots \\ M_{n-1} \otimes I_d - I_d \otimes M_{n-1}^{\mathsf{T}} \end{pmatrix}. \tag{9}$$

Its kernel is precisely $\mathrm{End}_{B_n}(U)$. Write $\mathcal{C}_5 = \mathcal{C}(S)$ for the 3600×900 ambient system and $\mathcal{C}_5^{\mathrm{red}} = \mathcal{C}(A)$ for the 1600×400 reduced system. Their entries belong to $\mathbb{Z}[q, q^{-1}, \tau]$.

We fix the minors by the following deterministic selection rule. Evaluate at $(q, \tau) = (2, 3)$ over $\mathbb{F}_{1009}$ and perform forward elimination, scanning columns from left to right. In each pivot column, choose the first nonzero row at or below the current pivot position, swap it into that position, normalize its pivot to one, and eliminate entries below. Record the original labels of the pivot rows and the pivot columns in their selection order. All initial labels are zero based, with rows ordered by generator and then by matrix entry. Let f_5 and g_5 be the determinants of the selected submatrices of the Laurent systems $\mathcal{C}_5$ and $\mathcal{C}_5^{\mathrm{red}}$, respectively. The certificate repository at `https://github.com/the-omega-institute/jtl-seeds-problem68` gives their full sparse Laurent matrices as well as the index lists.

Theorem 2. *The Laurent polynomial f_5 is nonzero. For every $(a, c) \in \mathcal{T}_5$ such that $f_5(e^{2\pi i a}, e^{2\pi i c}) \neq 0$, the representation $\mathcal{R}_5$ is irreducible over $\mathbb{C}$. This nonvanishing set is dense and open in $\mathcal{T}_5$. The same conclusions hold with g_5 in place of f_5.*

Proof. Exact elimination at $(2, 3)$ over $\mathbb{F}_{1009}$ gives $\operatorname{rank} \mathcal{C}_5 = 893$ and the selected determinant 992. Thus f_5 cannot be the zero integer Laurent polynomial. Its nonvanishing bounds the ambient commutant dimension above by $900 - 893 = 7$. On $\mathcal{T}_5$, [1, Proposition 5.7] gives

$$U_0 \cong \mathbf{1}^{\oplus 2} \oplus \mathrm{Bur}_5^{\mathrm{red}}(\tau) \oplus \mathrm{Bur}_5^{\mathrm{red}}(q) \oplus \mathcal{R}_5(q, \tau). \tag{10}$$

Consequently

$$\dim \mathrm{End}_{B_5}(U_0) \geq 4 + 1 + 1 + \dim \mathrm{End}_{B_5}(\mathcal{R}_5) \geq 7.$$

Equality forces $\mathrm{End}_{B_5}(\mathcal{R}_5) = \mathbb{C}I$. For $(a, c) \in \mathcal{T}_5$, choose $l = (1 - 5c - 6a)/2$ and $\lambda = e^{2\pi i l}$. Then $0 < l < 1 - 5c - 6a < 1$, so Negami's Theorem 5.3 supplies a positive-definite invariant form on the full 30-dimensional module, and Proposition 5.4(1) gives unitarizability without an irreducibility hypothesis. The braid matrices in (1) are independent of this choice of λ. Its restriction to $\mathcal{R}_5$ is positive definite. Any invariant subspace therefore has an invariant orthogonal complement. A proper nonzero invariant subspace would give a nonscalar commuting projection, proving the claimed irreducibility.

For the reduced system the exact rank is 399 and the selected determinant is 742 modulo 1009. Thus $g_5 \neq 0$; its nonvanishing forces a scalar reduced commutant directly, and the same positive-form argument applies. The rational witness $(2, 3)$ is used to prove nonvanishing, while the irreducibility conclusion is made on $\mathcal{T}_5$.

A nonzero Laurent polynomial cannot vanish on an open patch of the parameter torus. Its restriction is a real analytic finite Fourier sum; vanishing on a patch would force vanishing on the torus and then all coefficients to be zero. Hence each nonvanishing set is dense and open in $\mathcal{T}_5$. □

For comparison with [1, Proposition 5.10], the table gives the exact ranks and nullities at $(2, 3)$ over $\mathbb{F}_{1009}$ and the selected determinant residues there and at $(5, 7)$ over $\mathbb{F}_{1019}$.

Module	Dimension	Rank	Nullity	$\mathbb{F}_{1009}$	$\mathbb{F}_{1019}$
U_0, $n = 3$	12	137	7	786	18
U_0, $n = 4$	20	393	7	962	38
U_0, $n = 5$	30	893	7	992	393
$\mathcal{R}_5$	20	399	1	742	471

The $n = 3, 4$ rows reproduce the earlier certificate values; the $n = 5$ rows supply the new generic irreducibility certificates.

5 The diagonal splitting and factors of the minors

Write $[m]_q = 1 + q + \cdots + q^{m-1}$. An exact polynomial commuting operator gives the following splitting.

Lemma 3. *There is a matrix $K(q) \in \mathrm{Mat}_{20}(\mathbb{Z}[q])$, of entry degree at most eleven, such that*

$$[K, A_i(q, q)] = 0 \ (1 \le i \le 4), \qquad K^2 = q[6]_q^2 I, \qquad \operatorname{tr} K = 0, \tag{11}$$

and, in zero-based indices, $K_{1,0} = q[3]_q$.

Proof. Here is a finite specification of K using the matrices in (8). For each integer $v = 2, \ldots, 13$, solve the rational commutant equations for $A_i(v, v)$, together with $\operatorname{tr} K_v = 0$ and $(K_v)_{1,0} = v[3]_v$. These systems have unique solutions. Define

$$K(q) = \sum_{v=2}^{13} K_v \prod_{\substack{2 \le w \le 13 \\ w \ne v}} \frac{q - w}{v - w}. \tag{12}$$

The resulting entries have integer coefficients. Expansion of the four commutators, the square residual, the trace, and the specified entry verifies (11) coefficient by coefficient over $\mathbb{Z}[q, q^{-1}]$. The certificate records every entry of K. For completeness, the reduced commutant rank at each of the twelve interpolation points is 398 modulo 1019. This gives rational commutant dimension at most two; I and the verified $K(v)$ give the reverse bound. Since $K(v)$ has trace zero and prescribed nonzero entry, it is the unique normalized solution used in (12). Thus interpolation specifies the candidate, and the polynomial identities establish the assertion for all q. □

Theorem 4. *For every $0 < a < 1/11$, at $q = \tau = e^{2\pi i a}$,*

$$\mathcal{R}_5 = W_+ \oplus W_-, \qquad \dim W_+ = \dim W_- = 10.$$

These invariant subspaces are the images, in the coordinates of (8)*, of*

$$E_\pm = \frac{1}{2}\left(I \pm \frac{K(q)}{q^{1/2}[6]_q}\right), \qquad q^{1/2} = e^{\pi i a}. \tag{13}$$

Proof. The admissible diagonal in $\mathcal{T}_5$ is exactly $0 < a < 1/11$. Here $q \ne 1$ and $0 < 6a < 1$, so $[6]_q = (q^6 - 1)/(q - 1) \ne 0$. Equation (11) gives complementary idempotents: $E_\pm^2 = E_\pm$, $E_+ E_- = 0$, and $E_+ + E_- = I$. They commute with all generators. Their traces, hence their ranks, are ten. Their images are the required invariant summands. The isomorphism $\Psi|_{\ker D}$ transfers them to the augmentation kernel. □

Proposition 5. *Every minor of order 893 of $\mathcal{C}_5$ is divisible by $(\tau - q)^3$ in $\mathbb{Z}[q, q^{-1}, \tau]$. Every minor of order 399 of $\mathcal{C}_5^{\mathrm{red}}$ is divisible by $\tau - q$.*

Proof. On the admissible diagonal the two Burau summands in (10) are isomorphic, and scalar maps between the two copies give four independent endomorphisms. The two trivial copies give four more, and E_+, E_- give two. Hence the ambient commutant has dimension at least ten and the reduced commutant at least two. Their ranks are at most 890 and 398, respectively. The admissible arc is Zariski dense in the curve $\tau = q$, so these rank bounds hold over $\mathbb{Q}(q)$ after setting $\tau = q$.

Localize $\mathbb{Q}(q)[\tau]$ at $\tau - q$. A square submatrix of order 893 whose reduction has rank at most 890 has at least three invariant factors divisible by the uniformizer $\tau - q$. Its determinant is therefore divisible by the cube. The reduced argument uses orders 399 and 398. Finally, division by the monic polynomial $(\tau - q)^3$ over $\mathbb{Z}[q, q^{-1}]$ has a remainder of degree less than three in τ. Its image in $\mathbb{Q}(q)[\tau]$ is zero, so the remainder is zero. Division by $\tau - q$ gives the reduced assertion. □

6 Exact cyclotomic points and the remaining questions

Consider all 102 points

$$(a, c) = (j/84, k/84), \qquad j, k > 0, \qquad 6j + 5k < 84.$$

Set $\zeta = e^{2\pi i/84}$. The map $\zeta \mapsto 527 \in \mathbb{F}_{1009}$ defines a homomorphism $\mathbb{Z}[\zeta] \to \mathbb{F}_{1009}$, since $\Phi_{84}(527) = 0$ modulo 1009. The element 527 has order 84: its 84th power is one, while its powers $84/r$ for $r = 2, 3, 7$ are not one. Thus a nonzero reduced minor after this map certifies a nonzero cyclotomic minor in characteristic zero.

At all 95 off-diagonal points, g_5 has nonzero image modulo 1009, proving irreducibility over $\mathbb{C}$ by Theorem 2. At the seven diagonal points $j = k = 1, \ldots, 7$, the reduced modular rank is 398. The modular rank bounds the complex commutant dimension above by two; I and K give two independent complex endomorphisms. Its dimension is therefore exactly two. Complete reducibility makes this dimension the sum of squares of irreducible multiplicities. Together with the invariant $10 + 10$ splitting, this proves that W_+ and W_- are irreducible and inequivalent at these seven points. At $j = k = 1$ the ambient modular rank is 890, so its generic rank along the diagonal is exactly 890.

The verifier selects an order-398 minor of the full reduced commutant system at $q = \tau = 2$ over $\mathbb{F}_{1019}$, restricts it to $\tau = q$, and checks its determinant at $q = 2$ exactly over $\mathbb{Z}$. This gives a nonzero univariate Laurent polynomial $h(q)$. Whenever $h(q) \neq 0$, the reduced commutant has dimension two by the same I, K argument, so complete reducibility gives the corresponding conclusion for W_+ and W_-.

The exact finite computations supporting the paper are the commutant ranks and selected determinant residues, the Laurent coefficient identities in Lemma 3 and (7), and the cyclotomic reductions at the stated points. The Jordan obstruction is uniform in n, ℓ and the sector; it follows from the local construction, without a finite sector scan. For the Burau family, the certificates prove irreducibility on $\{f_5 \neq 0\} \cap \mathcal{T}_5$ and $\{g_5 \neq 0\} \cap \mathcal{T}_5$, and the polynomial projectors prove reducibility along the entire admissible diagonal.

The complete exceptional locus away from the diagonal, the residual factors and their exact multiplicities, and the image-closure and SU(20) questions in [1, Problem 6.8] remain open. A zero of one selected minor alone does not imply reducibility. Since $h(q)$ has only finitely many roots, W_+ and W_- are irreducible and inequivalent at all but finitely many admissible diagonal points; the complete exceptional diagonal set remains undetermined.

Data availability

The exact certificates and reproduction programs are available at `https://github.com/the-omega-institute/jtl-seeds-problem68`.

Use of AI tools

We used an AI orchestration system built on OpenAI's ChatGPT and Codex and Anthropic's Claude. AI agents searched for proofs and counterexamples, ran computations, and produced the Lean formalization of the ambient Jordan obstruction. Separate ChatGPT and Codex runs double-checked the arguments and computations; Claude coordinated the work. AI tools also assisted in preparing this manuscript.

References

[1] Haru Negami, *Quantum gates from the middle convolution of twisted Burau representations*, arXiv:2610.00293v1 (2026). `https://arxiv.org/abs/2610.00293v1`.